ECEN 164 HW 7

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DC gain = 35dB

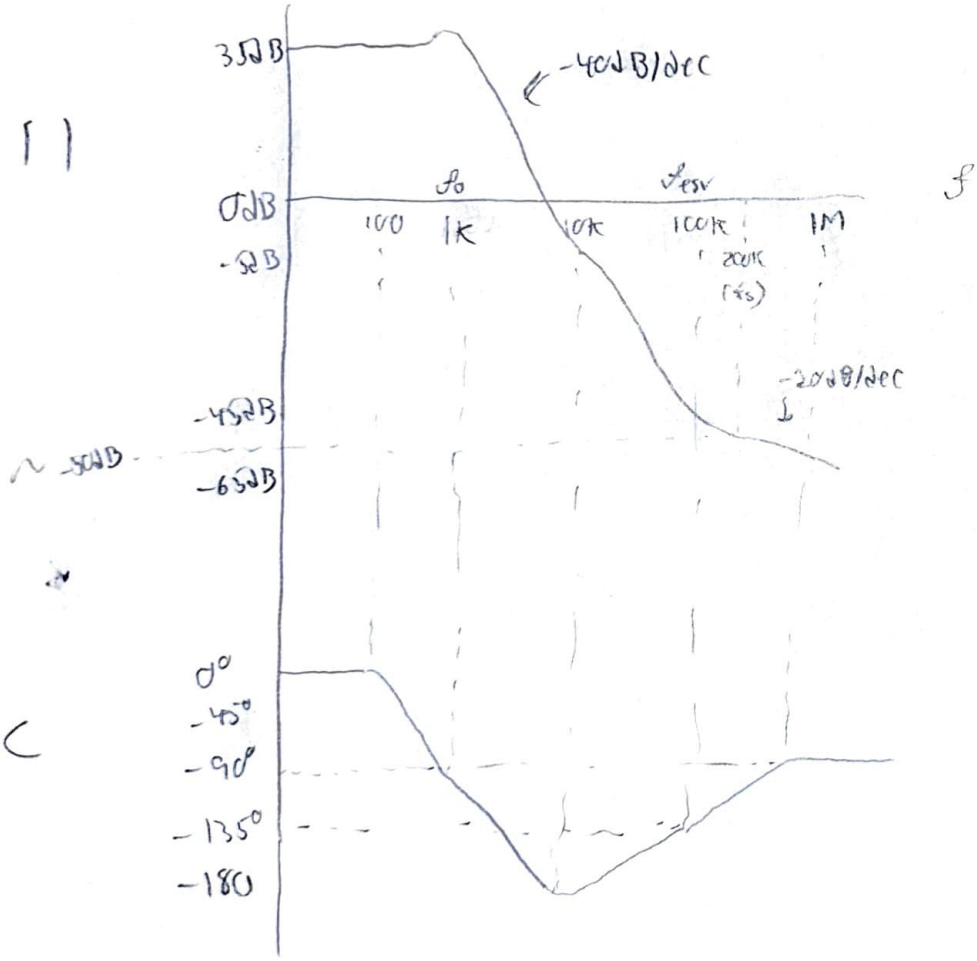
$L = 125\mu H$ $ESR = 7.95m\Omega$ $V_{peak} = 2V$ desired margin of 45 at $f_{co} = 10kHz$

$C = 200\mu F$ $f_s = 200kHz$

$f_o = \frac{1}{2\pi\sqrt{LC}} = 1006.584Hz$ $f_{esr} = \frac{1}{2\pi 200\mu F(7.95m\Omega)} = 100097.4485Hz$

a.

b. the f_s is rejected by about -50dB, which is really good.



c. The system briefly hits (or gets really close to) -180°. This is not good for stability. It may barely work in practice, but I say $\boxed{\text{not stable}}$

enough

d. filter + converter = -5dB at 10K

$G_{PWM} = 20\log(\frac{1}{V_{peak}}) = -6.02$

$-5 - 6.02 = -11.02$

$\boxed{G_{compensator} = 11.02dB}$

e.

$45 = -180 + \theta_{comp}$

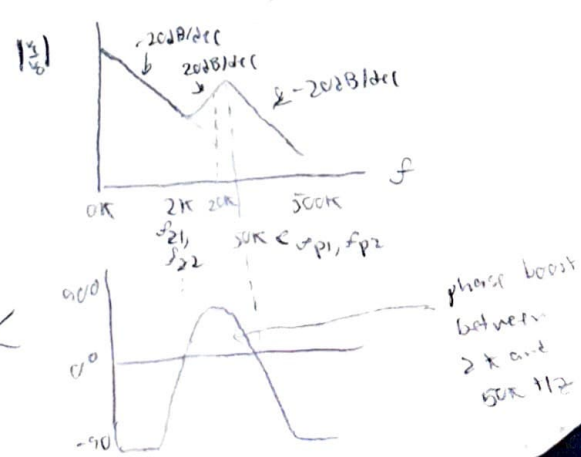
$\theta = 225$ type 3

$k = \tan\left(\frac{225-90}{4}\right)^2$

$k = 25.274$

$\boxed{f_z = \frac{f_{co}}{\sqrt{k}} = 1989.124Hz = f_{z1} = f_{z2} \quad f_{p0} = 0 \quad f_p = f_{co}\sqrt{k} = 50273.395Hz = f_{p1} = f_{p2}}$

f.



g. f_{esr} would decrease and we'd get less f_s rejection (more noise, -20dB/dec slope out 10x)

h. Yes. With a much smaller f_{esr} we'd now use a type 2 as the phase wouldn't get to -180.

i. High ESR = more noise, but more stable

low ESR = less noise, but more unstable